An Industrial Oriented Mini Project Report on

SAR Marine Surveillance: Marine Surveillance for Maritime objects and oil spills using GenAI

Submitted in Partial fulfillment of requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

In

COMPUTER SCIENCE AND ENGINEERING

By

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Under the guidance of

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
(AN AUTONOMOUS INSTITUTION)
Accredited by NBA & NAAC, Approved by AICTE, Affiliated to JNTUH.
Narayanaguda, Hyderabad, Telangana-29

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

CERTIFICATE

This is to certify that this is a bonafide record of the project report titled **"SAR Marine surveillance: Marine Surveillance for Maritime Objects and oil spills using GenAI"** which is being presented as the **Industrial Oriented Mini Project** report in **III year II Semester (KR23)** by

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C. MAHESH KIRAN	**24BD5A0509**

In partial fulfillment for the award of the degree of Bachelor of Technology in Computer Science and Engineering at Keshav Memorial Institute of Technology (An Autonomous Institution) Narayanaguda, affiliated to the Jawaharlal Nehru Technological University Hyderabad, Hyderabad

Faculty Supervisor	**Head of Department**
Mr. C. SUNDEEP BABU	**Ms. Priyanka Saxena**

Submitted for Viva Voce Examination held on ____________________

External Examiner

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- To be the fountainhead in producing highly skilled, globally competent engineers.
- Producing quality graduates trained in the latest software technologies and related tools and striving to make India a world leader in software products and services.

Mission

- To provide a learning environment that inculcates problem solving skills, professional, ethical responsibilities, lifelong learning through multi modal platforms and prepares students to become successful professionals.
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To be among the region's premier teaching and research Computer Science and Engineering departments producing globally competent and socially responsible graduates in the most conducive academic environment.

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PO1. Engineering Knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2. Problem Analysis: Identify formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3. Design/Development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4. Conduct Investigations of Complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5. Modern Tool Usage: Create select, and, apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6. The Engineer and Society: Apply reasoning informed by contextual knowledge to societal, health, safety. Legal und cultural issues and the consequent responsibilities relevant to professional engineering practice.

PO7. Environment and Sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts and demonstrate the knowledge of, and need for sustainable development.

PO8. Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9. Individual and Team Work: Function effectively as an individual, and as a member or leader in diverse teams and in multidisciplinary settings.

PO10. Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11. Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12. Life-Long Learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

PSO1: An ability to analyze the common business functions to design and develop appropriate Information Technology solutions for social upliftments.

PSO2: Shall have expertise on the evolving technologies like Python, Machine Learning, Deep learning, IOT, Data Science, Full stack development, Social Networks, Cyber Security, Mobile Apps, CRM, ERP, Big Data, etc.

PROGRAM EDUCATIONAL OBJECTIVES (PEOs)

PEO1: To imbibe analytical and professional skills for successful careers and create enthusiasts to pursue advance education supplementing their career growth.

PEO2: Graduates will solve real time problems design, develop and implement innovative ideas by applying their computer engineering principles.

PEO3: Graduates will develop necessary skillset for industry by imparting state of art technology in various areas of computer science engineering.

PEO4: Graduates will engage in lifelong learning and be able to work collaboratively exhibiting high level of professionalism.

PROJECT OUTCOMES

P1: Accurately detect vessel locations in SAR imagery using deep learning models.
P2: Segment and map environmental hazards like oil spills with TransUNet.
P3: Provide a responsive, high-resolution interactive visualization experience via a DeepZoom viewer.
P4: Ensure reliable and scalable data processing and serving using a FastAPI backend.

MAPPING PROJECT OUTCOMES WITH PROGRAM OUTCOMES

PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
P1	H	M	H		H		H	M	H	H	H	M
P2	M		H		H			H	H	H	H	M
P3	H	H	H		M	H	M	M	H		M	H
P4	M		L			M	H		M	H	M	H

L – LOW M –MEDIUM H– HIGH

PROJECT OUTCOMES MAPPING WITH PROGRAM SPECIFIC OUTCOMES

PSO	PSO1	PSO2
P1	H	M
P2	H	H
P3	H	H
P4	M	L

PROJECT OUTCOMES MAPPING WITH PROGRAM EDUCATIONAL OBJECTIVES

PEO	PEO1	PEO2	PEO3	PEO4
P1	H	H	M	H
P2	H	H	H	M
P3	H	M	H	M
P4	M	L	H	L

DECLARATION

We hereby declare that the results embodied in the dissertation entitled **“SAR Marine Surveillance: Marine Surveillance for Maritime objects and oil spills using GenAI**” has been carried out by us together during the academic year 2025-26 as a partial fulfillment of the award of the B.Tech degree in Computer Science and Engineering from Keshav Memorial Institute of Technology (An Autonomous Institution) Narayanaguda, affiliated to JNTUH. We have not submitted this report to any other university or organization for the award of any other degree.

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ACKNOWLEDGEMENT

We take this opportunity to thank all the people who have rendered their full support to our project work. We render our thanks to **Dr. B L Malleswari**, Principal who encouraged us to do the Project.

We are grateful to **Mr. Neil Gogte**, Founder & Director, **Mr. S. Nitin,** Director for facilitating all the amenities required for carrying out this project.

We express our sincere gratitude to **Ms. Deepa Ganu**, Director Academic for providing an excellent environment in the college.

We are also thankful to **Ms. Priyanka Saxena**, Head of the Department for providing us with time to make this project a success within the given schedule.

We are also thankful to our Faculty Supervisor **Mr. C. SUNDEEP BABU**, for his valuable guidance and encouragement given to us throughout the project work.

We would like to thank the entire CSE Department faculty, who helped us directly and indirectly in the completion of the project.

We sincerely thank our friends and family for their constant motivation during the project work.

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ABSTRACT

SAR Marine Surveillance: Marine Surveillance for Maritime objects and oil spills using GenAI

The SAR Marine Surveillance System is an end-to-end, AI-powered web platform designed to automatically detect ships and oil spills from large-scale Synthetic Aperture Radar (SAR) satellite imagery. SAR sensors operate in all weather conditions and at night, making them uniquely suited for continuous ocean monitoring — but their high-resolution data (160 MB+ per image) is difficult to analyze manually at scale.

This system addresses that challenge through a tiled inference pipeline: uploaded TIFF satellite images are converted into Deep Zoom Image (DZI) tile pyramids using PyVIPS, enabling both efficient browser-based viewing and GPU-memory-aware model inference. For ship detection, the Deformable DETR model (ResNet-50 backbone, fine-tuned on the SAR scope dataset combining HRSID and OPEN-SSDD sources) processes each 512×512 tile independently, maps detections to global pixel coordinates, and applies Non-Maximum Suppression across tile boundaries to produce final bounding boxes. For oil spill segmentation, the TransUNet architecture (ResNet-50 + 12-layer Vision Transformer encoder with U-Net decoder, trained on the Deep-SAR Oil Spill SOS dataset covering Gulf of Mexico and Persian Gulf events) predicts binary segmentation masks per tile, which are then stitched into a full-resolution output and re-converted to DZI for display.

The platform frontend, built with React and OpenSeadragon, presents synchronized dual viewers (original + detection overlay) for ship analysis and triple viewers (original, mask, color overlay) for oil spill analysis — all with deep-zoom pan/zoom synchronization. Annotorious.js-powered annotation workspaces allow analysts to manually draw bounding boxes and polygon masks, export them as JSON, and validate them through dedicated validator modules. Dual authentication (Firebase + Twilio Phone OTP) secures access, while an asynchronous job queue backed by Node.js, MongoDB, and FastAPI handles long-running inference without blocking the interface. The model server is containerized with Docker for portable deployment.

LIST OF FIGURES

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CHAPTER-1

1. INTRODUCTION

1.1 Purpose of the Project

The primary objective of the SAR Marine Surveillance System project is to design and implement an intelligent, unified maritime monitoring platform that bridges the gap between raw Synthetic Aperture Radar (SAR) satellite imagery and actionable maritime intelligence. This system serves as a comprehensive solution for automated ship detection, oil spill segmentation, and large-scale ocean surveillance by leveraging advanced AI models such as Deformable DETR and TransUNet. By integrating SAR image ingestion, Deep Zoom visualization, synchronized multi-view analysis, and human-in-the-loop annotation capabilities, the platform aims to enhance maritime security, environmental monitoring, disaster response, and illegal activity detection. The project creates a scalable and efficient ecosystem where maritime analysts can accurately monitor oceanic regions regardless of weather conditions or time of day, enabling faster decision-making, improved situational awareness, and reliable environmental protection.

1.2 Problem with Existing Systems

The current maritime surveillance landscape faces several critical limitations that reduce the efficiency and accessibility of SAR-based monitoring systems. The major challenges include:

- **Manual Analysis Limitation**: High-resolution SAR images cover vast ocean regions and contain numerous ships or oil spill patterns, making manual inspection slow, inconsistent, and impractical for large-scale monitoring.
- **Fragmented Workflows**: Existing solutions focus either on ship detection or oil spill segmentation, requiring organizations to use multiple disconnected tools and workflows for complete maritime analysis.
- **Hardware Constraints**: Large SAR TIFF images demand significant GPU memory, forcing many systems to downsample imagery and lose important spatial details required for detecting small vessels and fine spill regions.
- **Lack of Interactive Visualization**: Most academic and research tools provide outputs only in technical formats such as JSON or arrays.
- **Absence of Human Validation**: Current automated systems provide limited support for analyst-driven correction, annotation, and validation, reducing reliability.

1.3 Proposed System

The proposed SAR Marine Surveillance System is an AI-powered maritime monitoring platform that combines SAR satellite imagery processing, deep learning detection, and interactive visualization into a unified ecosystem. The system introduces:

- **Tiled Deep Learning Pipeline**: Large SAR TIFF images are converted into Deep Zoom Image (DZI) tiles, enabling efficient browser-based viewing and memory-optimized AI inference without losing high-resolution details.
- **Automated Ship Detection**: A fine-tuned Deformable DETR model detects ships of different scales from SAR imagery using multi-scale transformer attention and global coordinate mapping for accurate vessel localization.
- **Oil Spill Segmentation**: A TransUNet-based hybrid CNN-Transformer model generates precise binary masks for marine oil spills by combining spatial feature extraction with long-range contextual understanding
- **Asynchronous Processing Framework**: Detection tasks are executed asynchronously through a FastAPI background processing pipeline with MongoDB job tracking, allowing non-blocking uploads and real-time status monitoring.
- **Interactive Visualization & Annotation:** OpenSeadragon synchronized viewers and Annotorious.js provide deep-zoom visualization, AI overlay comparison, and human-in-the-loop annotation capabilities for validating and correcting model predictions.

1.4 Scope of the Project

The scope of the SAR Marine Surveillance System project encompasses the full-stack development of an AI-powered maritime monitoring platform tailored for SAR satellite image analysis, ship detection, oil spill segmentation, and analyst-driven validation workflows.

1.4.1 Functional Scope

- **Authentication & Security** Multi-mode secure authentication using Firebase (Email/Password, Google OAuth) and Phone OTP verification with Twilio Verify and JWT-based session management.
- **SAR Image Processing**: Upload and management of high-resolution SAR TIFF images categorized for ship detection or oil spill analysis, with automatic DZI tile generation using PyVIPS.
- **AI-Based Detection**: Automated asynchronous ship detection using Deformable DETR and

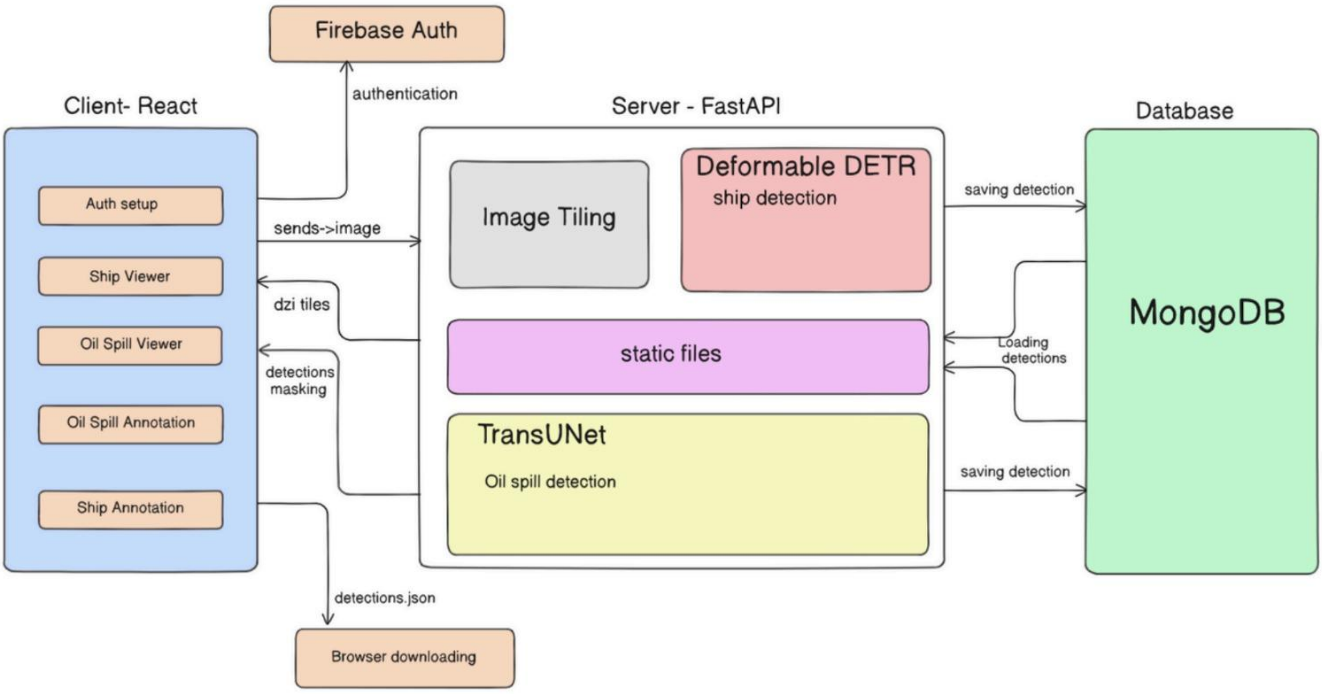
oil spill segmentation using TransUNet, with MongoDB-based result storage and full-resolution output generation.

- **Interactive Visualization**: Dual and triple synchronized OpenSeadragon viewers for side-by-side comparison of original imagery, detection overlays, segmentation masks, and color-highlighted outputs.
- **Annotation & Validation:** Human-in-the-loop annotation workspaces using Annotorious.js for drawing ship bounding boxes and oil spill polygon masks, including JSON export and validation utilities with statistical analysis.

1.4.2 Technical & Operational Scope

- **Tiled Inference Pipeline** Memory-efficient tiled inference architecture supporting large SAR imagery through DZI pyramids, global coordinate mapping, and mask stitching techniques.
- **Asynchronous Job Management**: Background detection pipeline with job status tracking (queued, running, completed, failed) using FastAPI tasks, webhook callbacks, and frontend polling mechanisms.
- **Data Persistence & Integration**: MongoDB integration for storing uploaded images, AI detections, annotations, and processing metadata for scalable maritime monitoring operations.
- **Cross-Platform Accessibility:** Browser-based responsive interface accessible to maritime analysts and non-technical users without requiring specialized GIS or Python expertise.

1.5 Architecture Diagram



Architecture Diagram

CHAPTER-2

2. LITERATURE SURVEY

2.1 Introduction

The maritime surveillance sector is rapidly evolving from manual SAR image interpretation toward AI-driven intelligent monitoring systems capable of automated ship detection and oil spill analysis. With the increasing availability of high-resolution Synthetic Aperture Radar (SAR) satellite imagery from platforms such as Sentinel-1A and ALOS, there is a growing need for scalable systems that can process large oceanic datasets under all-weather and day-night conditions. This project evaluates and integrates advanced deep learning architectures, tiled image processing pipelines, and interactive visualization technologies to develop a robust maritime surveillance platform based on Deformable DETR for ship detection and TransUNet for oil spill segmentation.

2.2 Key Methodologies

Ship Detection using Deformable DETR

The system utilizes Deformable DETR, a transformer-based object detection model optimized for SAR imagery. Multi-scale deformable attention enables efficient detection of ships of varying sizes while reducing computational complexity. The model is trained on SAR scope datasets (HRSID + OPEN-SSDD) using ResNet-50 feature extraction and achieves high detection accuracy for maritime vessel localization.

Oil Spill Segmentation using TransUNet

Oil spill detection is performed using TransUNet, a hybrid CNN-Transformer segmentation architecture combining ResNet-50 and Vision Transformer encoders. The model captures both local spatial details and long-range contextual dependencies to generate accurate binary oil spill masks from SAR imagery. Training on the Deep-SAR SOS dataset achieves strong segmentation performance across multiple spill scenarios.

SAR Datasets and Image Processing

The platform utilizes SARscope datasets for ship detection and the Deep-SAR SOS dataset for oil spill segmentation. Large SAR TIFF images are converted into Deep Zoom Image (DZI) pyramids using PyVIPS, enabling efficient tile-based processing and browser visualization without loading full-resolution images into memory.

Visualization and Annotation

OpenSeadragon enables interactive deep-zoom visualization of gigapixel SAR images, while Annotorious.js provides bounding box and polygon-based annotation tools for human validation and correction of AI detections. Exported annotations are compatible with model training and evaluation workflows.

Challenges

SAR imagery presents challenges such as speckle noise, low contrast, oil spill look-alike phenomena, limited labeled datasets, and GPU memory constraints for high-resolution images. These issues require tiled inference pipelines, robust deep learning architectures, and analyst-driven validation mechanisms.

Applications

The system supports maritime security, illegal fishing detection, oil spill monitoring, vessel traffic analysis, and AI-assisted maritime research. Its scalable architecture enables rapid environmental monitoring and automated surveillance across large oceanic regions.

Research Gaps and Future Directions

Future improvements include multi-temporal vessel tracking, multi-class ship classification, AIS integration for dark ship detection, real-time satellite data ingestion, and federated learning for collaborative maritime intelligence systems.

CHAPTER-3

3. SOFTWARE REQUIREMENT SPECIFICATION

3.1 Introduction to SRS

The Software Requirements Specification (SRS) serves as the primary technical foundation for the SAR Marine Surveillance System , detailing the functional and non-functional requirements of the platform. It acts as a comprehensive reference for developers, testers, and stakeholders to ensure the accurate implementation of AI-driven ship detection, oil spill segmentation, and SAR image analysis through a unified digital interface. This document bridges the gap between the conceptual vision of intelligent maritime surveillance and its practical implementation using React, Node.js, MongoDB, FastAPI, Deformable DETR, and TransUNet models.

3.2 Role of SRS

The SRS functions as the foundational blueprint for the SAR Marine Surveillance System , providing a clear framework for implementing AI-based ship detection, oil spill segmentation, image management, and annotation workflows. It minimizes development ambiguity by defining the system architecture, functional requirements, API structure, and database design while also serving as the validation benchmark for testing and future system enhancements.

3.3 Requirements Specification Document

This section formalizes the technical requirements of the SAR Marine Surveillance System , including AI-based ship detection, oil spill segmentation, SAR image management, annotation workflows, and deep-zoom visualization. It provides a structured specification of functional requirements, non-functional constraints, performance standards, and software technologies used throughout the platform architecture.

3.4 Functional Requirements

The SAR Marine Surveillance System must support the following core functionalities to ensure efficient maritime monitoring and AI-assisted analysis:

- **Secure Authentication**: The system shall support Firebase authentication (email/password, Google OAuth) and Twilio OTP-based login with JWT session management.

- **SAR Image Processing**: Users must be able to upload SAR TIFF images categorized for ship or oil spill analysis with automatic DZI tile generation using PyVIPS.
- **AI-Based Detection**: The platform shall support asynchronous ship detection using Deformable DETR and oil spill segmentation using TransUNet with job status tracking.
- **Interactive Visualization**: The system must provide synchronized OpenSeadragon viewers for comparing original SAR imagery with detection overlays and segmentation masks.
- **Annotation & Validation**: Analysts must be able to create, export, and validate ship bounding boxes and oil spill polygon annotations using Annotorious.js tools.

3.5 Non-Functional Requirements

Beyond core functionalities, the system must maintain strict operational standards for reliability, scalability, and usability:

- **Data Security**: All protected routes must require authenticated access, and OTP sessions must use secure JWT-based authentication mechanisms.
- **System Scalability**: The tiled inference pipeline must efficiently process very large SAR images without exhausting GPU or system memory resources.
- **Reliability**: Failed AI detection jobs must return proper error handling and status updates to avoid silent system failures.
- **Maintainability**: Backend services and API routes must remain modular to support future AI model upgrades and additional maritime detection features.
- **Usability**: The interface must provide responsive deep-zoom visualization with real-time status updates and theme support for maritime analysts.

3.6 Performance Requirements

3.6.1 Ship Detection Performance

The Deformable DETR model must perform inference on a single 512×512 SAR image tile in less than 2 seconds on CPU hardware. This ensures efficient large-scale ship detection across tiled SAR satellite imagery.

3.6.2 Oil Spill Segmentation Performance

The TransUNet segmentation model must generate predictions for a 224×224 SAR tile within 1.5 seconds on CPU systems. This allows rapid processing and reconstruction of full-resolution oil spill masks.

3.6.3 DZI Generation Performance

The system must convert large SAR TIFF images (approximately 160 MB) into Deep Zoom Image (DZI) pyramids using PyVIPS in under 60 seconds to support efficient browser-based visualization.

3.6.4 Viewer Response Time

The OpenSeadragon viewer must begin rendering image tiles within 1 second after image selection, ensuring smooth and responsive deep-zoom interaction for maritime analysts.

3.6.5 Error Handling and Recovery

The platform must gracefully handle failed uploads, inference errors, and interrupted jobs by providing clear status updates and recoverable workflows for analysts.

3.7 Software Requirements

3.7.1 System Software

- **Operating Systems**: Compatible with modern web browsers on Windows, Linux, and macOS platforms for browser-based maritime surveillance and analysis.
- **Server Environment**: Linux-based environments are used for hosting the Node.js backend, MongoDB database, and FastAPI AI model-serving infrastructure.

3.7.2 Development Software

- **Backend**: Express.js is utilized for REST API development, file upload handling, asynchronous job routing, and integration with AI services.
- **Frontend**: React 18 with Vite 5 is used to build a responsive, component-based user interface with high-performance development and rendering capabilities.
- **Database**: MongoDB with Mongoose is employed for storing SAR image metadata, job tracking records, AI detections, and annotation data.

3.7.3 Supporting Libraries

- **Visualization & Annotation**: OpenSeadragon and Annotorious.js are used for deep-zoom SAR image viewing and annotation of ship bounding boxes and oil spill polygons.
- **Authentication**: Firebase Authentication and Twilio Verify API provide secure email, Google OAuth, and phone OTP-based login systems.
- **Image Processing**: PyVIPS (libvips) is used for efficient TIFF-to-DZI conversion and tiled SAR image processing.

3.8 Hardware Requirements

3.8.1 Minimum Requirements (Development & Training)

- **Storage**: Minimum 100 GB storage space for SARscope, Deep-SAR SOS datasets, generated tiles, and model checkpoints.
- **RAM**: At least 16 GB system RAM required for dataset loading and tiled inference workflows.

3.8.2 Recommended Requirements (Client-Side)

- **Browser**: Latest versions of Google Chrome 120+ or Firefox 115+ with JavaScript enabled for OpenSeadragon visualization.
- **Display**: High-resolution monitor recommended for detailed maritime image analysis and annotation tasks.
- **Internet**: Stable broadband connection for uploading large TIFF SAR images and loading DZI tiles efficiently.

3.8.3 Server Requirements

- **CPU**: Minimum 4-core x86_64 processor for backend services and CPU-based AI inference.
- **RAM:** At least 16 GB RAM to support concurrent detection requests, model loading, and DZI generation.
- **GPU:** NVIDIA GPU with CUDA 11.8+ for accelerated deep learning inference and faster processing.

CHAPTER-4

4. SYSTEM DESIGN

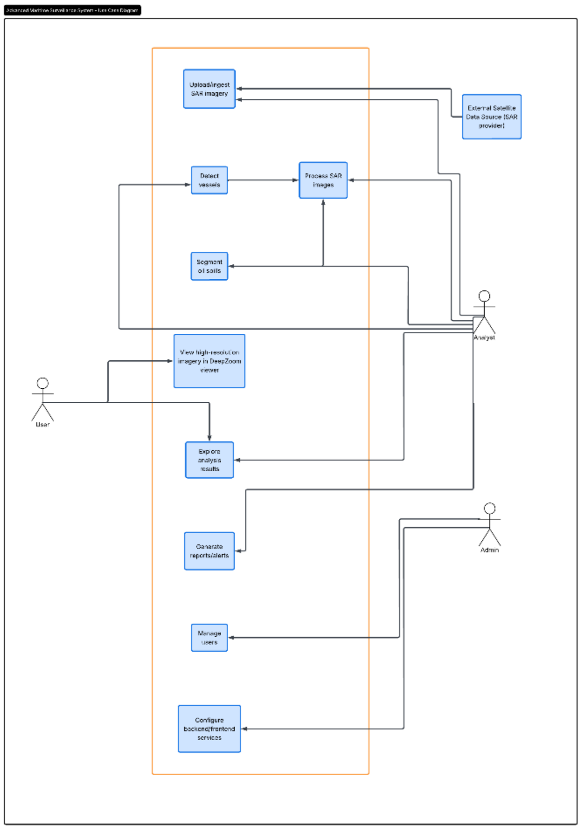
4.1. Introduction to UML

The Unified Modeling Language (UML) is a standardized modeling language used to visualize, design, and document the structure and behavior of software systems. It provides a clear and organized way to represent system components, data flow, and interactions among modules and users.

4.2. UML Diagrams

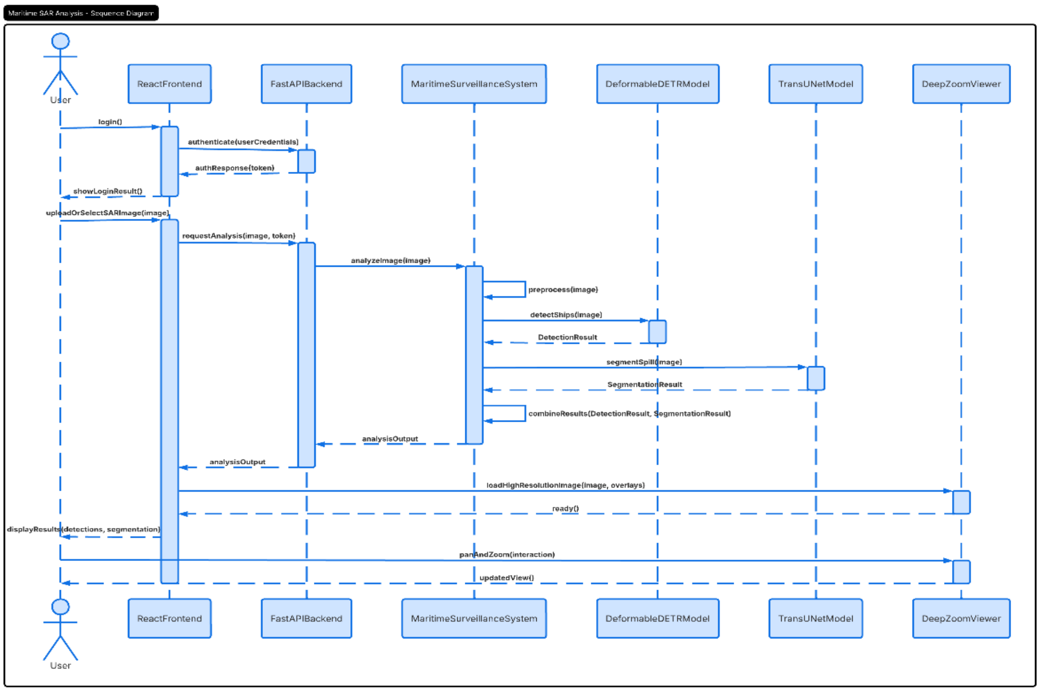
This section outlines the key UML diagrams created for the SAR Marnie Surveillance, illustrating the functional flow, component interactions, and overall architecture.

4.2.1. Use Case Diagram



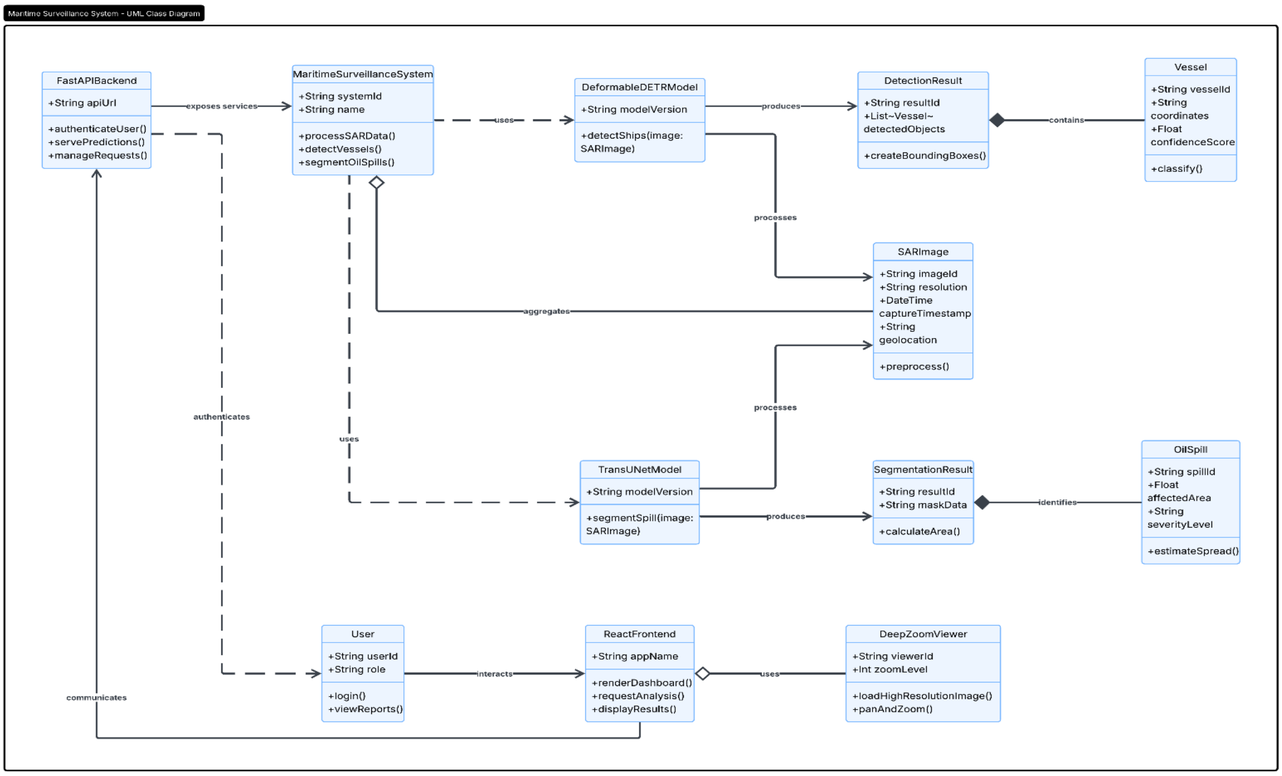
4.2.1 Use Case Diagram

4.2.2 Sequence Diagram



4.2.2 Sequence Diagram

4.2.3 Class Diagram



4.2.3 Class Diagram

4.3 Technologies Used

The SAR Marine Surveillance System is developed using a full-stack MERN-based architecture integrated with advanced AI frameworks, SAR image processing libraries, and deep-zoom visualization technologies to support maritime monitoring and automated satellite image analysis.

Core Technologies:

- React:

  Used for building a responsive, component-based frontend interface with fast rendering and efficient state management for synchronized SAR image viewers.

- OpenSeadragon:

  Deep Zoom Image (DZI) viewer used for rendering gigapixel SAR satellite imagery with synchronized viewport support and overlay visualization.

- Express.js + Node.js:

  Backend framework used for REST API development, asynchronous job routing, file uploads, and webhook communication between frontend and AI services.

- MongoDB + Mongoose:

  NoSQL database used for storing SAR image metadata, asynchronous detection jobs, annotations, and AI-generated outputs.

- FastAPI + Uvicorn

  Python-based asynchronous model server used for hosting Deformable DETR and TransUNet inference pipelines with Background Task support.

- PyTorch:

  Deep learning framework utilized for training and inference of transformer-based ship detection and oil spill segmentation models.

- PyVIPS (libvips):

  High-performance image processing library used for converting large SAR TIFF images into Deep Zoom Image (DZI) tile pyramids.

- Firebase Authentication & Twilio Verify:

  Used for secure multi-provider authentication including email/password login, Google OAuth, and phone OTP verification.

- Docker:

  Containerization platform used for reproducible deployment of the FastAPI model-serving environment and AI dependencies.

CHAPTER-5

5. IMPLEMENTATION

5.1 Setting up Connections

The implementation phase begins with establishing a scalable three-tier architecture connecting the React frontend, Node.js middleware backend, and FastAPI-based AI model server. The React frontend is initialized using Vite and integrated with Firebase Authentication for secure analyst access, while React Router manages protected navigation across detection and annotation modules. The Express.js backend configures REST APIs, TIFF image uploads, MongoDB connectivity, and static DZI tile serving to support asynchronous maritime analysis workflows.

Simultaneously, the FastAPI Python model server initializes the Deformable DETR and TransUNet deep learning models during startup, ensuring efficient inference without repeated model loading overhead. Shared directories for uploads, DZI tiles, and AI outputs are configured across all layers, enabling seamless communication between the frontend visualization system, backend APIs, and AI inference pipeline. GPU acceleration through PyTorch CUDA support is enabled when available, while CPU fallback ensures compatibility across different deployment environments.

5.2 Coding the Logic

The system logic is architected to handle large-scale SAR image processing, AI inference workflows, and asynchronous maritime monitoring operations:

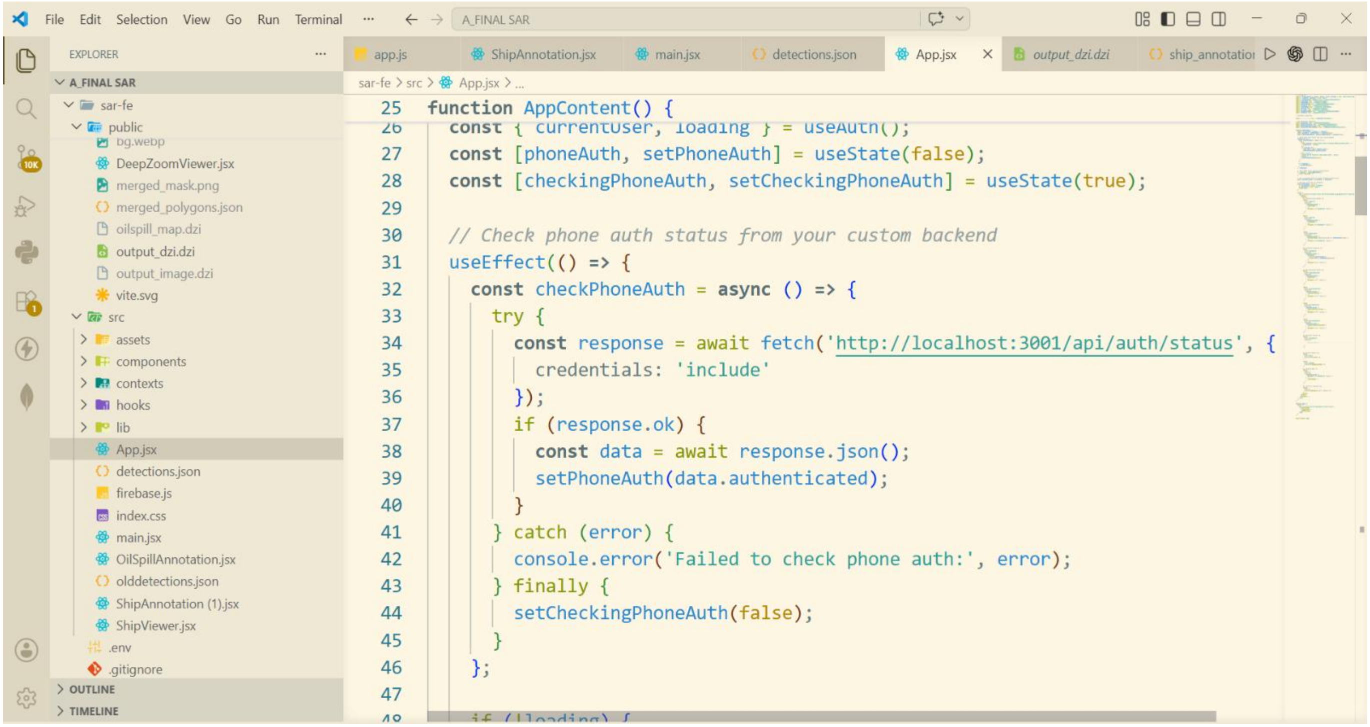
• **Tiled Inference Pipeline**: The core image-processing logic uses PyVIPS to convert large SAR TIFF images into Deep Zoom Image (DZI) tiles, enabling memory-efficient processing and browser-based visualization.

• **AI Detection Workflow**: Ship detection logic is implemented using Deformable DETR with global coordinate mapping and Non-Maximum Suppression (NMS), while oil spill segmentation uses TransUNet with tile-wise prediction and full-resolution mask stitching.

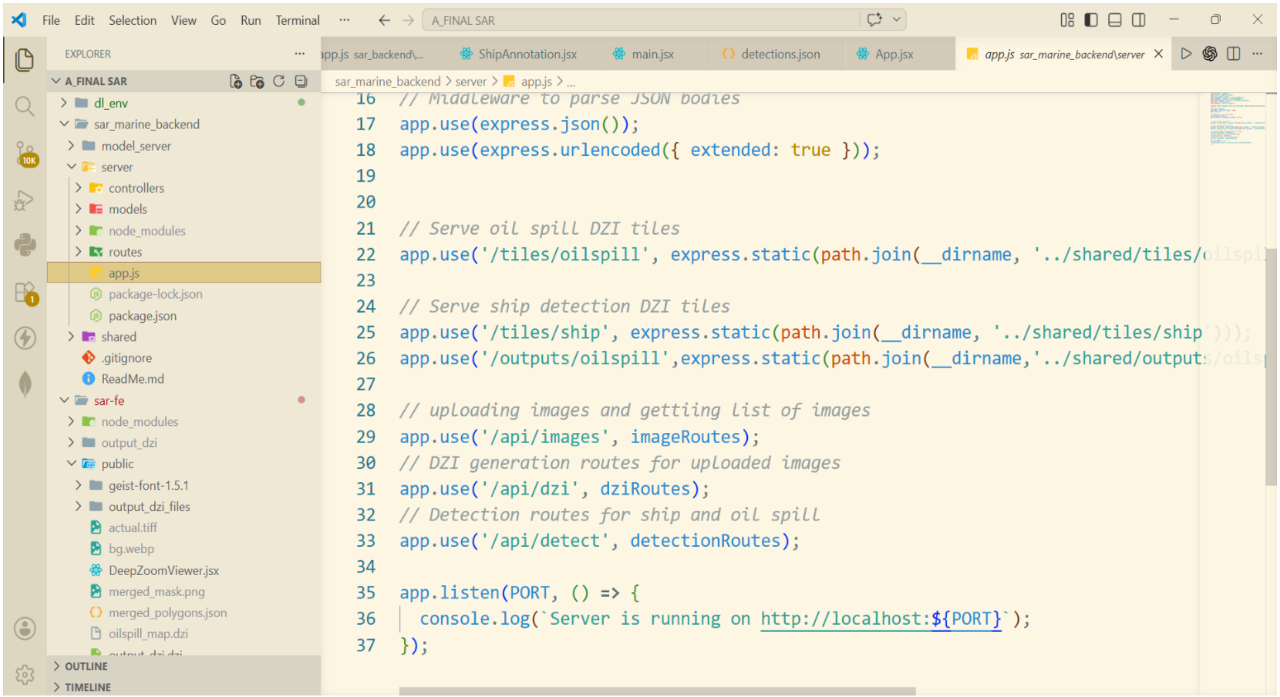
• **Authentication Flow**: Secure analyst authentication is implemented using Firebase Authentication, JWT session management, and Twilio OTP verification for protected access to maritime analysis modules.

• Asynchronous Job Processing: Detection requests are handled asynchronously through FastAPI Background Tasks, MongoDB job tracking, webhook callbacks, and frontend polling for real-time status updates.

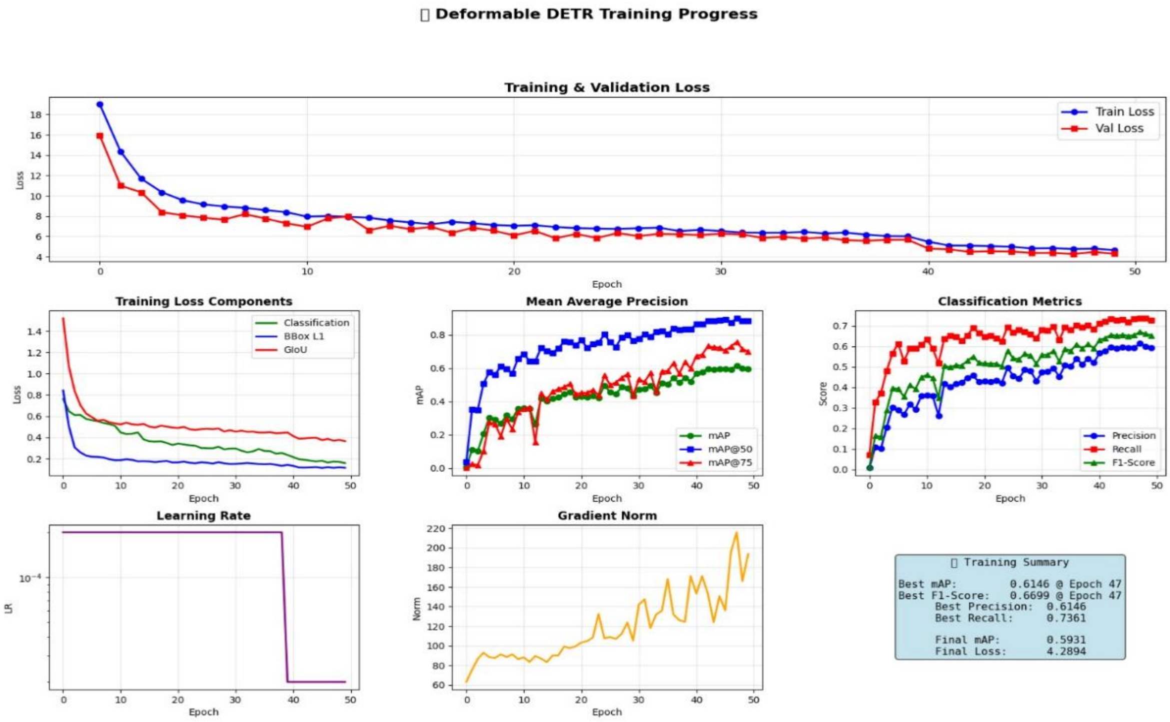
• **Interactive Visualization & Annotation**: OpenSeadragon and Annotorious.js handle synchronized SAR image viewing, detection overlays, and analyst-driven annotation workflows for validation and correction of AI outputs.



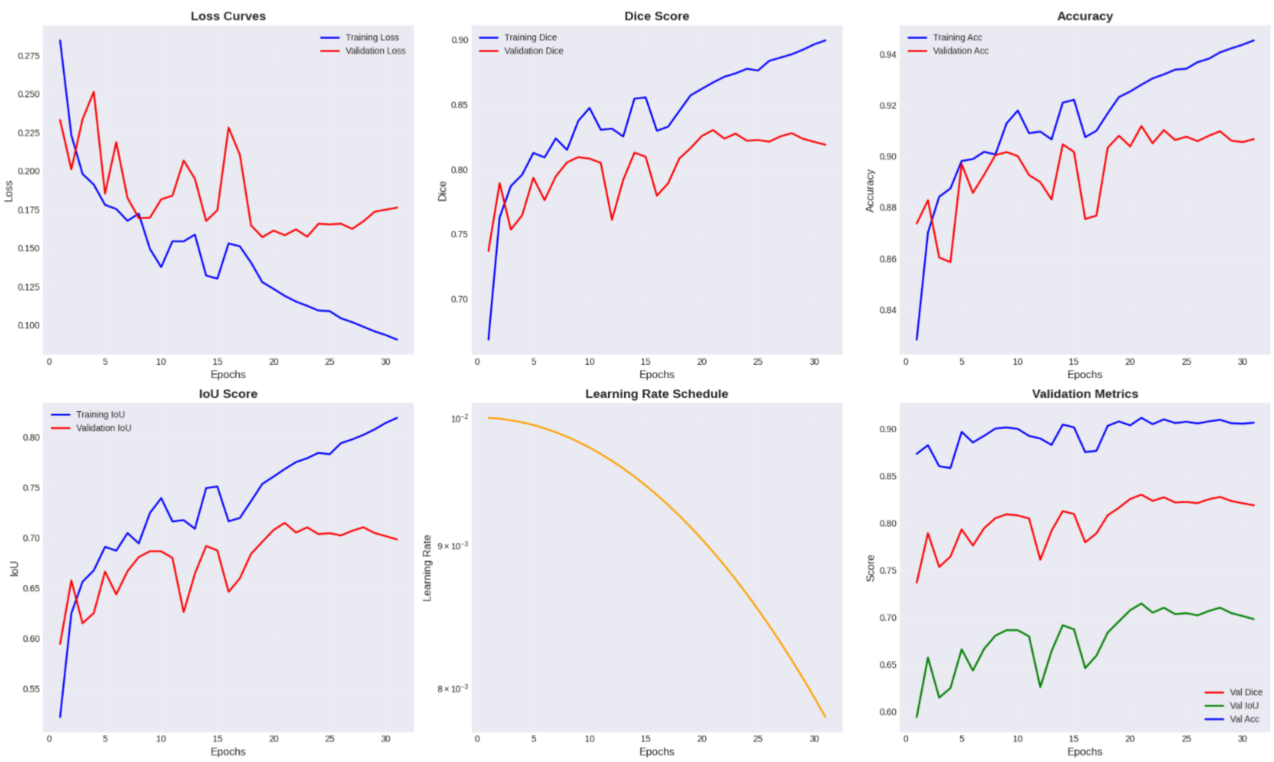
Frontend App.jsx



Backend app.js



TransUNet Model Metrics



Deformable DETR Model Metrics

5.3 Connecting the Dashboard

The administrative dashboard acts as the central monitoring and management interface for the SAR Marine Surveillance System.

• **Centralized Monitoring**: The dashboard connects directly to MongoDB to provide real-time monitoring of uploaded SAR images, AI detection jobs, annotation records, and system activity.

• **Detection Management**: Administrators and analysts can track asynchronous ship detection and oil spill segmentation tasks, monitor job status (queued, running, completed, failed), and review generated AI outputs.

• **Visualization Control**: The dashboard integrates OpenSeadragon viewers to display synchronized SAR imagery, ship detection overlays, oil spill masks, and annotation layers for detailed maritime analysis.

• **Annotation & Validation Management**: Analysts can validate AI-generated detections, upload annotation JSON files, review bounding boxes and polygon masks, and export corrected datasets for future model improvement.

• **Analytics & Reporting**: The platform provides statistical summaries such as ship count, detection density, oil spill coverage area, and annotation metrics to support maritime surveillance and environmental monitoring operations.

5.4 Screenshots

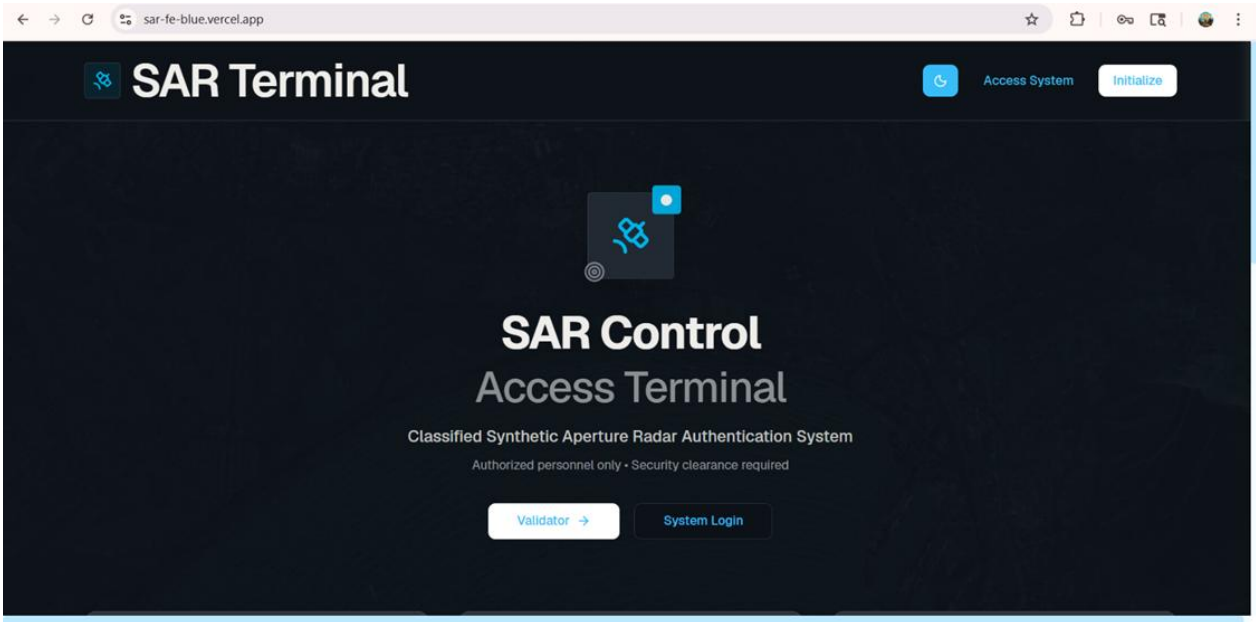
This section provides visual documentation of the backend and architectural components:

This section provides visual documentation of the SAR Marine Surveillance System ’s frontend, backend, AI inference pipeline, and visualization components:

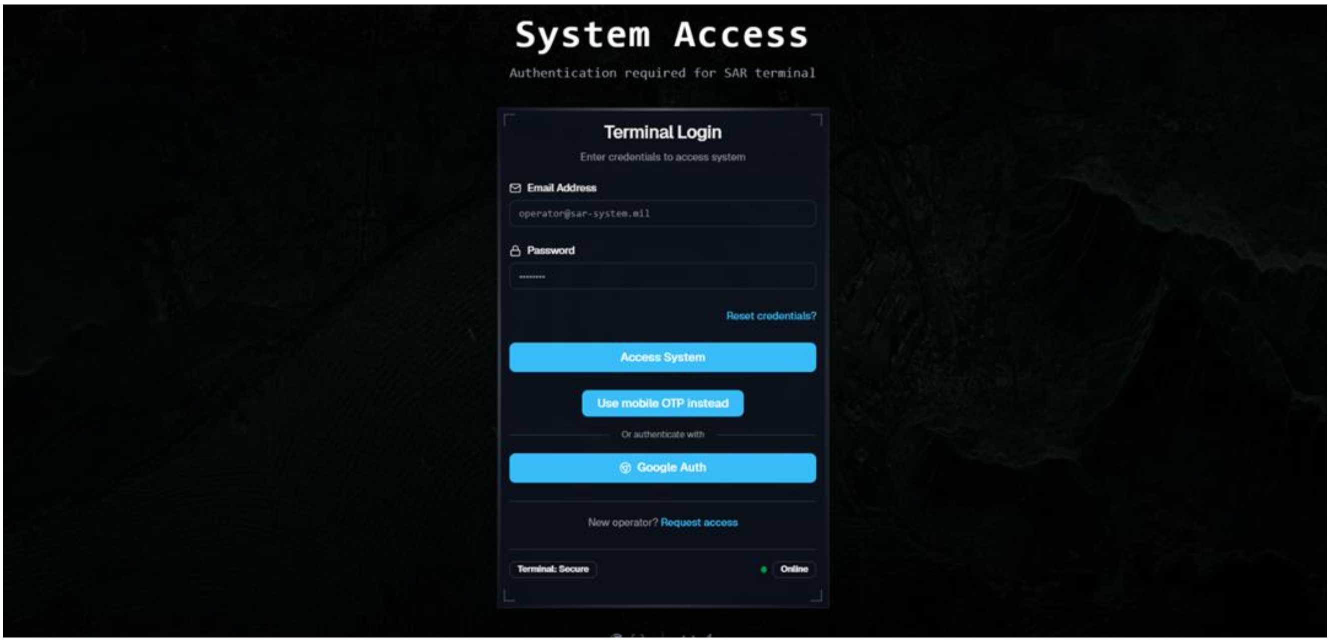
• Frontend Interface: Screenshots of the React-based user interface including authentication pages, SAR image upload modules, and synchronized OpenSeadragon viewers for ship and oil spill analysis.

• AI Detection Outputs: Visuals demonstrating ship detection bounding boxes generated by Deformable DETR and oil spill segmentation masks produced by TransUNet on SAR imagery.

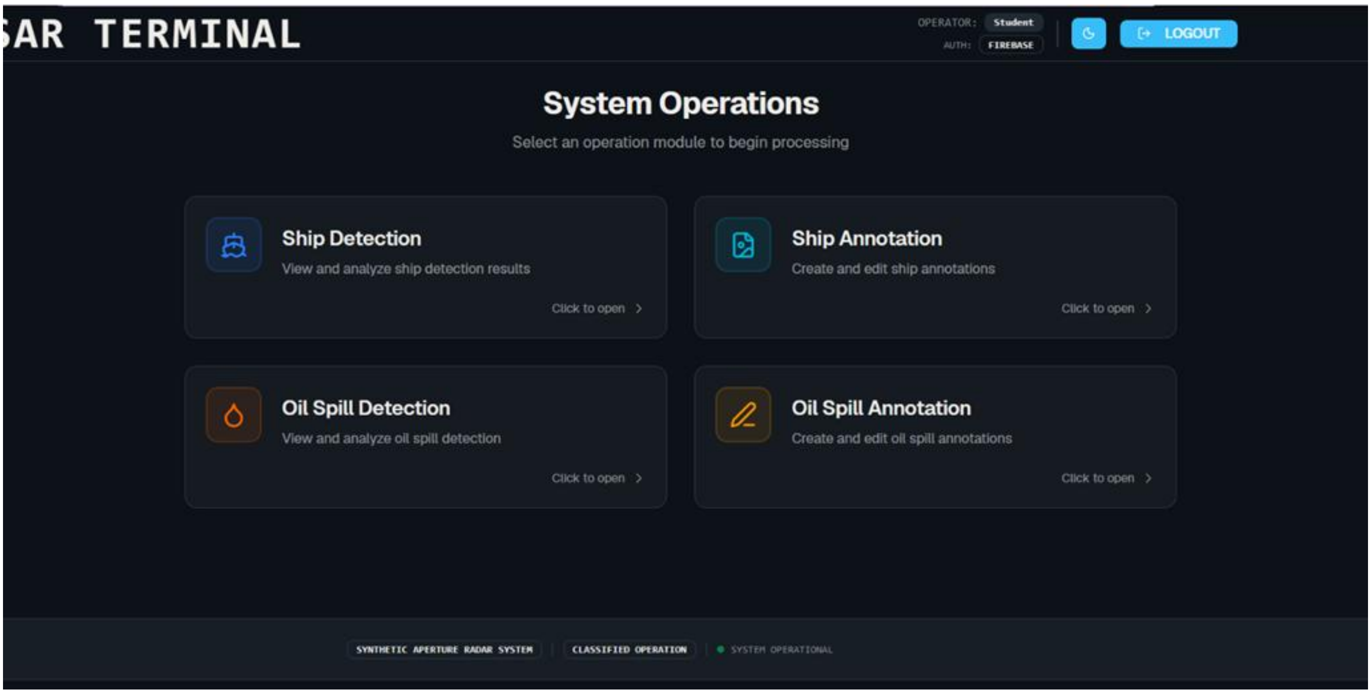
• Annotation & Validation Modules: Screenshots of Annotorious.js-based annotation workspaces showing bounding box drawing, polygon mask annotation, and JSON validation interfaces.



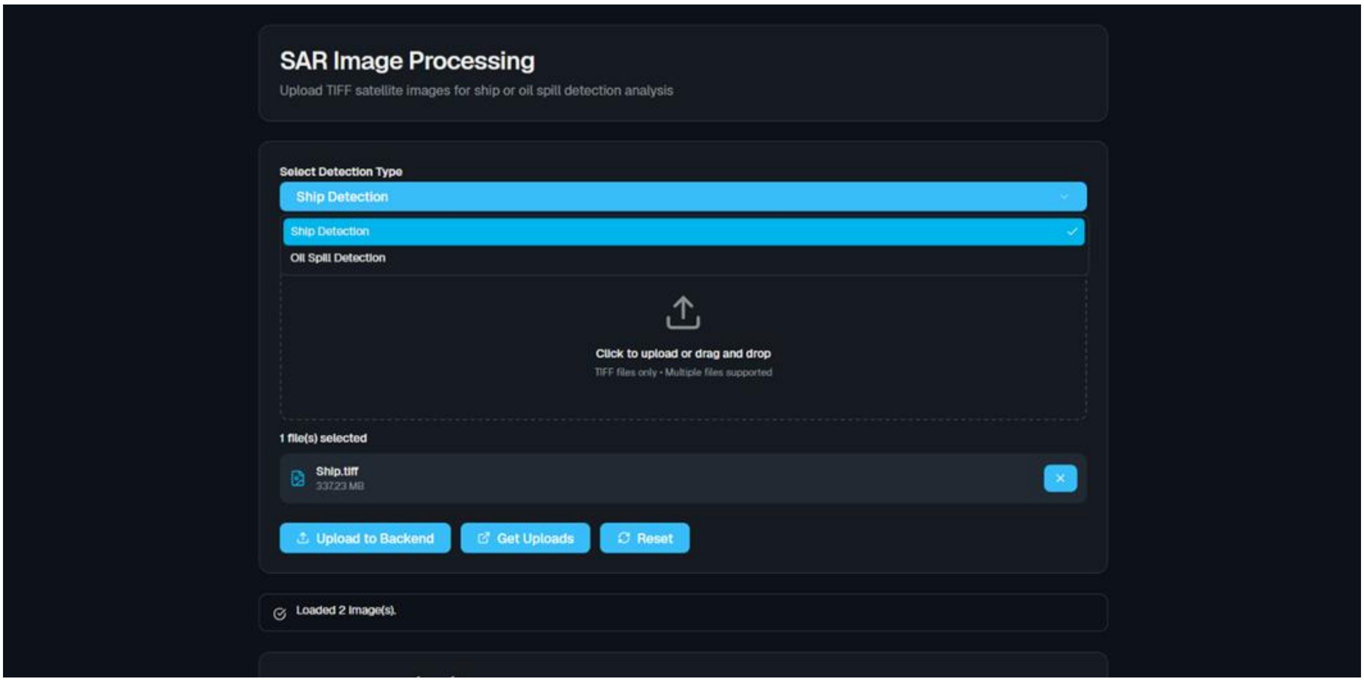
Landing Page



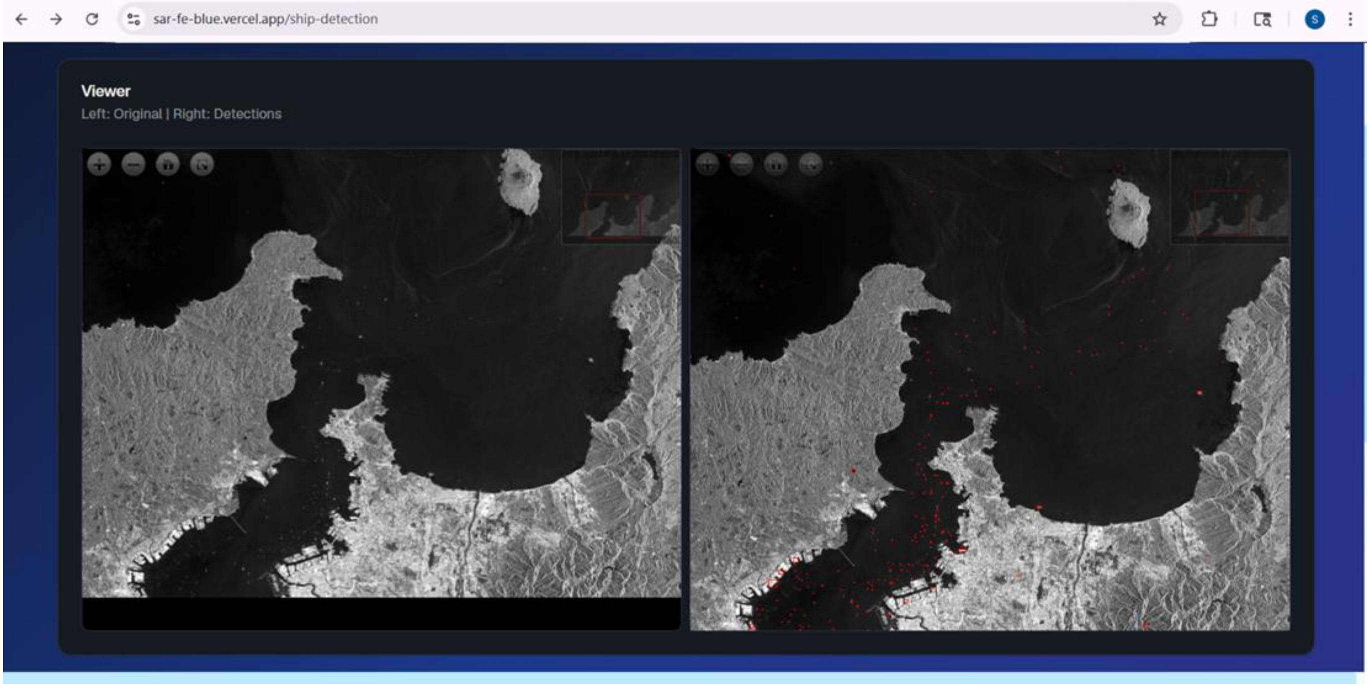
Login Page



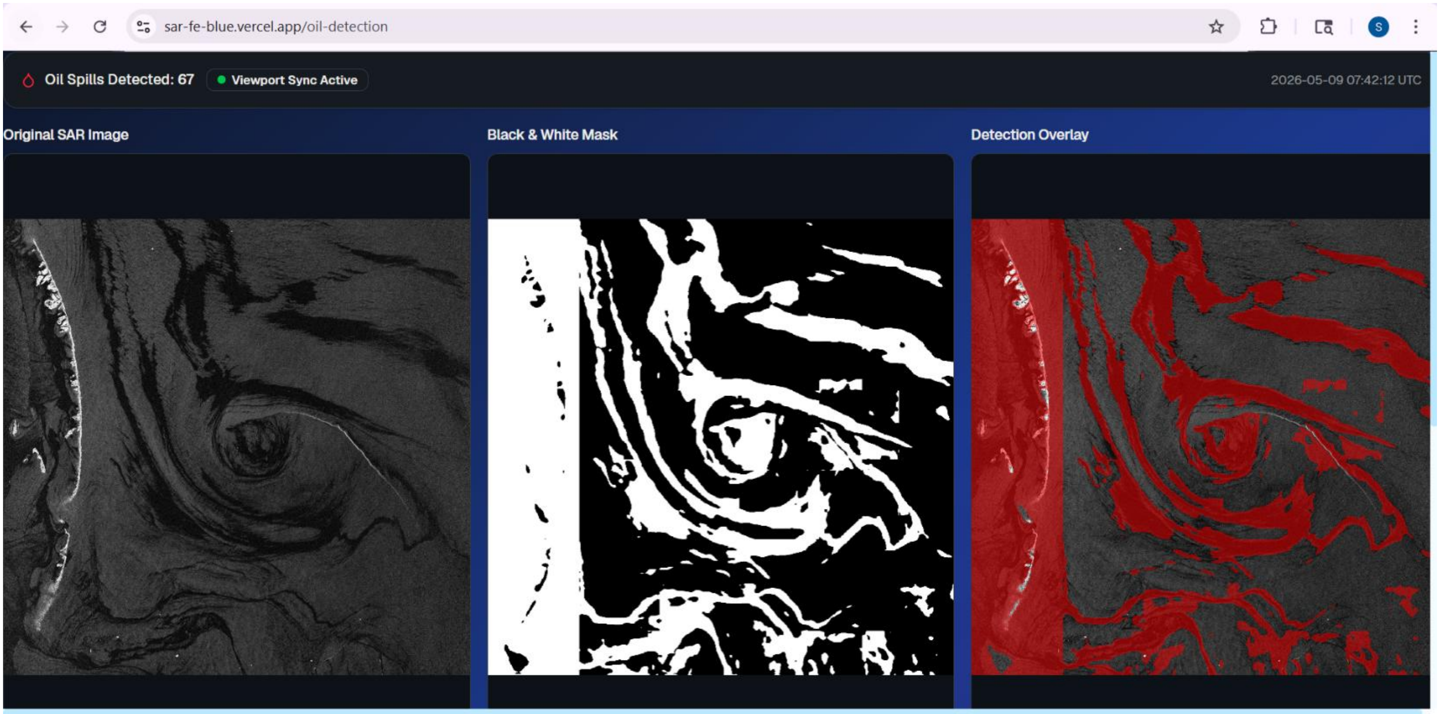
Landing Page after Logging in



TIFF IMAGE Upload Page

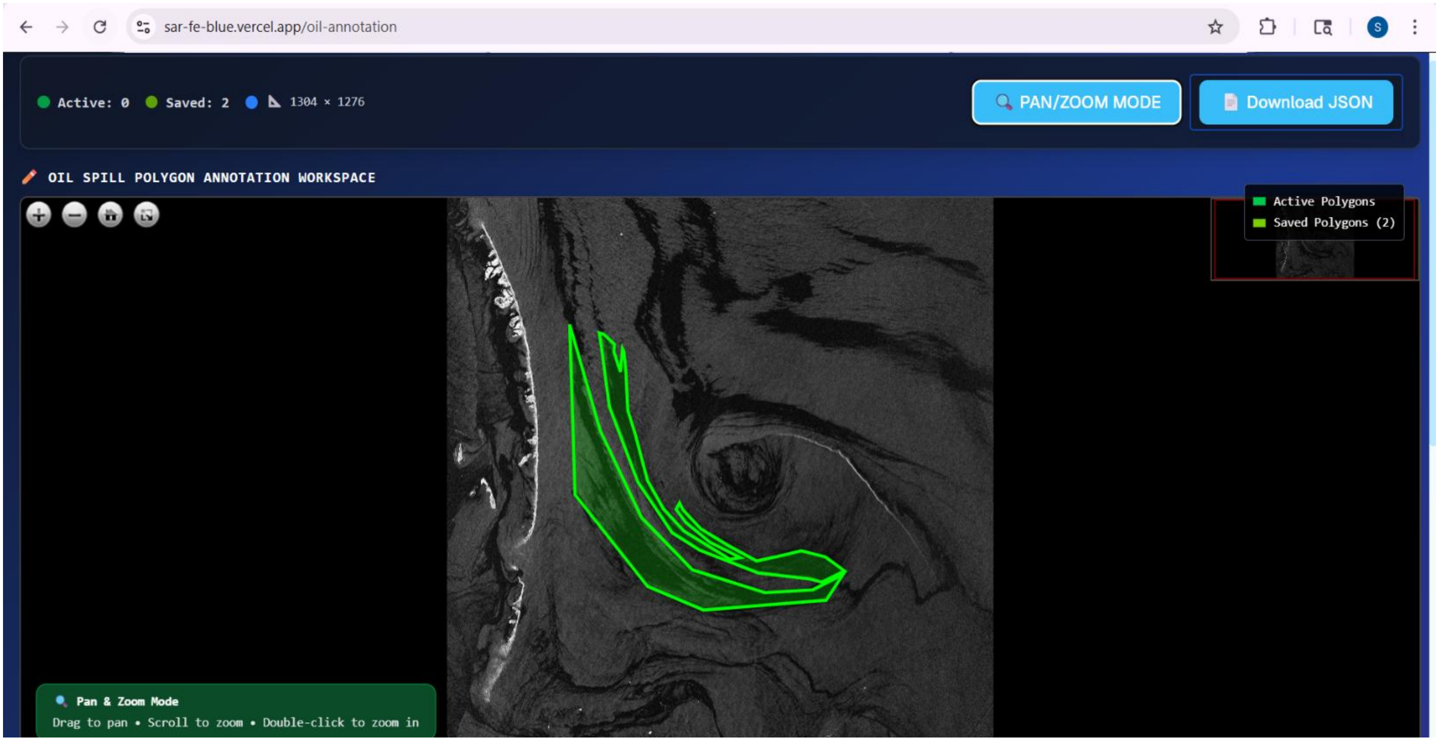


Ship Detection Page



OIL Detection Page

Ship Annotation Page



OIL Spill Annotation Page

CHAPTER-6

6. SOFTWARE TESTING

6.1 Introduction

Software testing for the SAR Marine Surveillance System ensures that all three tiers — React frontend, Node.js backend, and Python model server — function correctly, reliably, and securely. Given the AI-intensive nature of the system, testing encompasses both traditional software testing (functional, integration, system, security) and ML-specific evaluation (model accuracy metrics on held-out test sets). Testing verified that the tiled inference pipeline correctly maps tile-local detections to global coordinates, that DZI viewers synchronize correctly, and that the asynchronous job queue handles all state transitions reliably.

6.1.1 Testing Objectives

The primary goal of testing is to validate that the SAR Marine Surveillance System is functionally accurate, operationally reliable, and technically secure for real-world maritime monitoring applications. Key objectives include:

• Validation of Core Functions: Ensuring that SAR image upload, DZI generation, ship detection, oil spill segmentation, annotation, and validation workflows operate correctly without failure.

• Performance Benchmarking: Verifying that the tiled inference pipeline, asynchronous job processing system, and OpenSeadragon visualization maintain efficient response times and stable operation for large SAR imagery.

• Security Assurance: Confirming that Firebase authentication, Twilio OTP verification, and JWT-based session management correctly restrict unauthorized access and protect analyst operations.

• AI Model Validation: Ensuring that the Deformable DETR and TransUNet models achieve the required accuracy targets for ship detection and oil spill segmentation respectively.

6.1.2 Testing Strategies

A multi-layered testing strategy is used to validate the functionality, integration, and performance of the SAR Marine Surveillance System.

• Unit Testing: Individual Python inference functions, tiled coordinate mapping, mask stitching, and Node.js backend routes are tested independently using Jest and Supertest.

• Integration Testing: End-to-end workflows including TIFF upload, DZI generation, AI inference, webhook callbacks, MongoDB updates, and frontend retrieval are validated using real SAR test images and Postman API testing.

• System Testing: Complete user workflows such as authentication, image upload, detection, annotation, and visualization are tested across integrated frontend and backend components.

• Model Evaluation: Deformable DETR and TransUNet models are evaluated on SARscope and Deep-SAR SOS datasets using metrics such as AP@0.5, Dice Score, IoU, and Accuracy.

6.1.3 System Evaluation

The system evaluation phase assesses how effectively the SAR Marine Surveillance System performs automated ship detection, oil spill segmentation, and large-scale SAR image analysis. This includes evaluating the accuracy of the Deformable DETR and TransUNet models, the efficiency of the tiled inference pipeline, and the reliability of asynchronous detection, visualization, and annotation workflows.

6.1.4 Testing New System

Testing the SAR Marine Surveillance System involves simulating real-world maritime monitoring and SAR image analysis scenarios:

• Tiled Inference Testing: Processing large SAR TIFF images to verify efficient DZI generation, tile-wise AI inference, and accurate global coordinate mapping without boundary artifacts.

• Cross-Platform Testing: Ensuring consistent visualization, authentication, and annotation behavior across modern web browsers such as Chrome and Firefox.

• Asynchronous Workflow Testing: Validating webhook-based job processing, MongoDB status updates, and real-time frontend polling during active ship detection and oil spill segmentation tasks.

6.2 Test Cases

Test Case ID	Feature	Description	Expected Result	Outcome
TC-P01	Authentication	Register a new analyst account using email/password authentication.	System creates account successfully and redirects user to the dashboard.	All Passed
TC-P02	Phone OTP Verification	Verify login using phone number and OTP authentication.	System verifies OTP, creates JWT session, and grants secure access.	All Passed
TC-P03	SAR Image Upload	Upload a SAR TIFF image for ship detection analysis.	TIFF image uploads successfully, and imageId is generated in MongoDB.	All Passed
TC-P04	DZI Generation	Generate Deep Zoom Image (DZI) tiles from uploaded SAR TIFF.	DZI manifest and tile pyramid are created successfully for OpenSeadragon viewing.	Partially passed
TC-P05	Ship Detection	Trigger Deformable DETR inference on uploaded SAR imagery.	System detects ships accurately and renders bounding boxes in synchronized viewers.	All Passed
TC-P06	Oil Spill Segmentation	Run TransUNet segmentation on oil spill SAR imagery.	Binary oil spill mask is generated, stitched, and displayed correctly in triple viewer mode.	Partially failed
TC-P07	Annotation & Validation	Export annotation JSON and validate it using the validation module.	Bounding boxes and polygon annotations render correctly with statistical summaries.	All Passed

CONCLUSION

The SAR Marine Surveillance System successfully addresses the growing need for intelligent maritime monitoring by transforming raw Synthetic Aperture Radar (SAR) satellite imagery into actionable maritime intelligence through an AI-powered web platform. By integrating advanced transformer-based deep learning architectures such as Deformable DETR for ship detection and TransUNet for oil spill segmentation, the system achieves high detection accuracy while overcoming the computational challenges associated with processing large SAR satellite images through an efficient tiled inference pipeline.

The implementation of a three-tier architecture consisting of a React frontend, Node.js backend, and FastAPI-based AI model server enables a complete end-to-end analyst workflow, including SAR image upload, asynchronous AI inference, synchronized deep-zoom visualization, annotation, and validation. The integration of OpenSeadragon and Annotorious.js provides an interactive environment for maritime analysts, while Firebase authentication and Twilio OTP verification ensure secure platform access.

Furthermore, the system demonstrates the practical applicability of transformer-based AI models in real-world maritime surveillance, environmental monitoring, illegal fishing detection, and coast guard operations. By successfully implementing and validating all functional and non-functional requirements, the project establishes a scalable, secure, and deployable framework capable of supporting next-generation AI-assisted maritime intelligence systems.

FUTURE ENHANCEMENTS

While the current version of the SAR Marine Surveillance System provides a strong foundation for AI-driven maritime monitoring, several advanced enhancements can further improve its operational capabilities and real-world deployment potential:

• **Geographic Coordinate Integration:** Future versions can integrate GDAL-based GeoTIFF metadata extraction to convert pixel-level detections into latitude and longitude coordinates. This would allow ship and oil spill detections to be visualized on interactive GIS platforms such as Openseadragon.

• **Multi-Class Ship Classification:** The Deformable DETR model can be extended to classify multiple vessel categories such as cargo ships, fishing boats, tankers, patrol vessels, and military ships using advanced SAR datasets like FUSAR-Ship and OpenSARShip 2.0. This enhancement would support maritime threat analysis and illegal fishing detection.

• **Real-Time Satellite Data Ingestion:** Integration with ESA Copernicus and NASA Earthdata APIs could enable automated ingestion of newly available Sentinel-1 SAR imagery for continuous monitoring of maritime zones and time-series analysis.

• **AIS-Based Dark Ship Detection:** Future systems may cross-reference SAR-detected vessels with Automatic Identification System (AIS) broadcasts to identify "dark ships" that disable AIS transmissions, a common indicator of smuggling, illegal fishing, or suspicious maritime activity.

• **Expanded SAR Dataset Integration:** Incorporating additional datasets such as LS-SSDD-v1.0, Sentinel-1 Oil Spill datasets, and Eastern Mediterranean Oil Slick datasets would improve model robustness, especially for small ship detection and diverse oil spill morphologies.

• **Automated Alert and Notification System:** Real-time push notifications, email alerts, and SMS-based warnings triggered by suspicious vessel activity or large oil spill detections could transform the platform into a proactive 24/7 maritime monitoring solution.

• Federated Learning for Secure Collaboration: Federated learning frameworks can enable multiple maritime organizations to collaboratively improve AI models without sharing sensitive SAR imagery, ensuring privacy-preserving distributed model updates.

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